

Project Initialization and Planning Phase

Date	
Team ID	Kalash suhag
Project Name	Analyzing Mental Health in Student Ecosystem
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Student Counselor	Identify students who require psychological support	It is difficult to understand the relationship between stress, depression, and academic performance.	there is no single visual platform showing all important mental health indicators	challenged in providing early and effective counseling support
PS-2	A Student Counselor	Identify students who require psychological support	It is difficult to understand the relationship between stress, depression, and academic performance	There is no single visual platform showing all important mental health indicators	Challenged in providing early and effective counseling support

Initial Project Planning Template

Date	
Team ID	Kalash suhag
Project Name	Analysing Mental Health in Student Ecosystemth
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Preparation	USN-1	As a user, I can register for the application by entering my email, and password, and confirming my password.	2	High	Kalash suhag	15.7.26	15.7.26
Sprint-1	Data Analysis	USN-2	As a user, I will receive a confirmation email once I have registered for the application	1	High	Kalash suhag	15.7.26	15.7.26
Sprint-2	Visualisation Development	USN-3	As a user, I can register for the application through Facebook	2	Low	Kalash suhag	16.7.26	16.7.26

Sprint-1	Dashboard	USN-4	As a user, I can register for the application through Gmail	2	Medium	Kalash suhag	16.7.26	16.7.26
Sprint-1	Story Creation and Publhing	USN-5	As a user, I can log into the application by entering email & password	1	High	Kalash suhag	17.7.26	17.7.26

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Project Proposal (Proposed Solution) template

This project proposal outlines a solution to analyze students' mental health and academic performance using an interactive Tableau dashboard. The solution aims to transform raw student mental health data into meaningful visual insights that help students, educators, and counselors make informed decisions.

Project Overview	
Objective	To develop an interactive Mental Health Analytics Dashboard using Tableau that visualizes the relationship between study hours, stress levels, depression, academic performance, and demographic factors to support better mental health awareness and decision-making.

Scope	The project includes collecting and cleaning the mental health dataset, creating interactive visualizations in Tableau, developing a dashboard and story to present key insights, and generating reports for analysis. The project excludes real-time data collection, predictive machine learning models, and multi-user authentication systems.
Problem Statement	
Description	Educational institutions collect large amounts of student data related to stress, depression, study habits, and academic performance. However, this information is often stored in spreadsheets, making it difficult to identify patterns and students who may require psychological support. There is a lack of a centralized visual platform that combines these important mental health indicators for easy analysis.
Impact	An interactive dashboard enables educators and student counselors to quickly identify trends in mental health and academic performance, allowing timely interventions and better support for students. It also helps students understand the factors affecting their well-being and academic success.
Proposed Solution	
Approach	The solution involves cleaning and preprocessing the student mental health dataset, importing it into Tableau, creating interactive charts such as bubble charts, bar charts, treemaps, heatmaps, lollipop charts, and KPI cards, and combining them into an interactive dashboard and Tableau Story. Filters for age, gender, and study hours allow users to explore the data dynamically.
Key Features	• Interactive Dashboard: Visualizes study hours, stress levels, depression, academic performance, and demographic information. • Multiple Visualizations: Bubble chart, bar chart, treemap, heatmap, lollipop chart, KPI cards, and dashboard filters. • Student Insights: Identifies relationships between mental health indicators and academic performance.

	• Counselor Support: Helps counselors identify students who may need psychological assistance. • Dynamic Filtering: Users can filter data by gender, age, and study hours for deeper analysis. • Tableau Story: Presents findings in a structured and easy-to-understand narrative.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU Specifications, number of cores	Intel core i5/i7 Processor
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Data Visualization Platform	Tableau Public 2026
Libraries	Built-in Tableau features and functions	Tableau calculated fields, filters, parameters, dashboard actions
Development Environment	Development and version control	Tableau Public 2026, Microsoft Excel 2021, Windows 11
Data		
Data	Source, size, format	Student mental health dataset (CSV format, approximately 1000 student records)

Data Collection and Preprocessing Phase

Date	
Team ID	Kalash suhag
Project Title	Analysing Mental Health in Student Ecosystem
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
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Project Overview	The Mental Health Analytics Dashboard is an interactive Tableau project designed to analyze the relationship between students' mental health and academic performance. The dashboard visualizes indicators such as study hours, stress levels, depression, academic performance, age, gender, and heart rate variability to support students, educators, and counselors in understanding mental health trends.
Data Collection Plan	Student mental health data was collected from a publicly available dataset containing demographic information, study habits, stress levels, depression scores, academic performance, and physiological indicators. The dataset was imported into Microsoft Excel for data cleaning and preprocessing before being loaded into Tableau for visualization and dashboard creation.
Raw Data Sources Identified	1. Student Mental Health Dataset: Contains information on age, gender, study hours, stress levels, depression levels, academic performance, and heart rate variability. 2. Processed Dataset: Cleaned CSV file with missing values handled, duplicate records removed, and data formatted for Tableau analysis.

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Student Mental Health Dataset	https://drive.google.com/file/d/190Qmq27LeZZ_nWriCp3Obl7ys_5otEsp/view?usp=share_link	CSV	317	Public

	contain ing study hours, stress, depress ion, academ ic perfor mance, age, gender, and heart rate variabil ity.			K B	
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Data Collection and Preprocessing Phase

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Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Student Mental Health Dataset	Missing values in study hours, stress level, and depression columns.	High	Identify missing values and remove or replace them using appropriate methods to maintain data accuracy.
Student Mental Health Dataset	Duplicate student records.	Moderate	Remove duplicate entries to avoid incorrect analysis and repeated data.
Student Mental Health Dataset	Inconsistent text formatting (e.g., "Male", "male", "MALE") in categorical fields.	Moderate	Standardize text by converting values to a consistent format and removing extra spaces.
Student Mental Health Dataset	Invalid or inconsistent numerical values (e.g., negative study hours or stress scores outside the valid range).	High	Validate numerical fields and correct or remove invalid records.
Student Mental Health Dataset	Different data formats for numerical and categorical columns.	Low	Convert all columns to the appropriate data types before importing into Tableau.
Student Mental Health Dataset	Outliers in study hours or academic performance values.	Medium	Review outliers and retain only valid observations or remove erroneous values after verification.

Data Collection and Preprocessing Phase

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Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	A comprehensive student mental health dataset containing demographic details (Age, Gender), academic information (Study Hours, Academic Performance), mental health indicators (Stress Level, Depression Level), and physiological metrics (Heart Rate Variability). The dataset is used to analyze the relationship between mental health and academic performance.
Data Cleaning	Missing Values: Identified and handled missing values by removing incomplete records or replacing them where appropriate. Duplicates: Removed duplicate student records to maintain data integrity. Error Correction: Standardized text formatting by removing extra spaces, correcting inconsistent capitalization, and ensuring consistent category names (e.g., Male/Female).
Data Transformation	Calculated Fields: Created calculated measures such as average stress level, average depression score, and average study hours. Aggregations: Used Tableau aggregations to calculate averages, counts, and percentages for different student groups. Filtering: Removed invalid or unrealistic values and applied filters based on age, gender, and study hours for focused analysis.
Data Type Conversion	Verified that Age, Study Hours, Stress Level, Depression Level, Academic Performance, and Heart Rate Variability are stored as numeric values, while Gender and other categorical attributes are stored as text dimensions for accurate visualization in Tableau.

Column Splitting and Merging	Renamed columns for better readability and consistency. Combined related attributes where required and ensured each variable was stored in a separate column to enable efficient filtering and analysis in Tableau.
Data Modeling	Connected the cleaned dataset within Tableau and created calculated fields, KPIs, and relationships between demographic variables and mental health indicators to support interactive dashboard visualizations.
Save Processed Data	Exported the cleaned and standardized dataset as a CSV/Excel file and imported it into Tableau Desktop for dashboard creation, story development, and report generation.

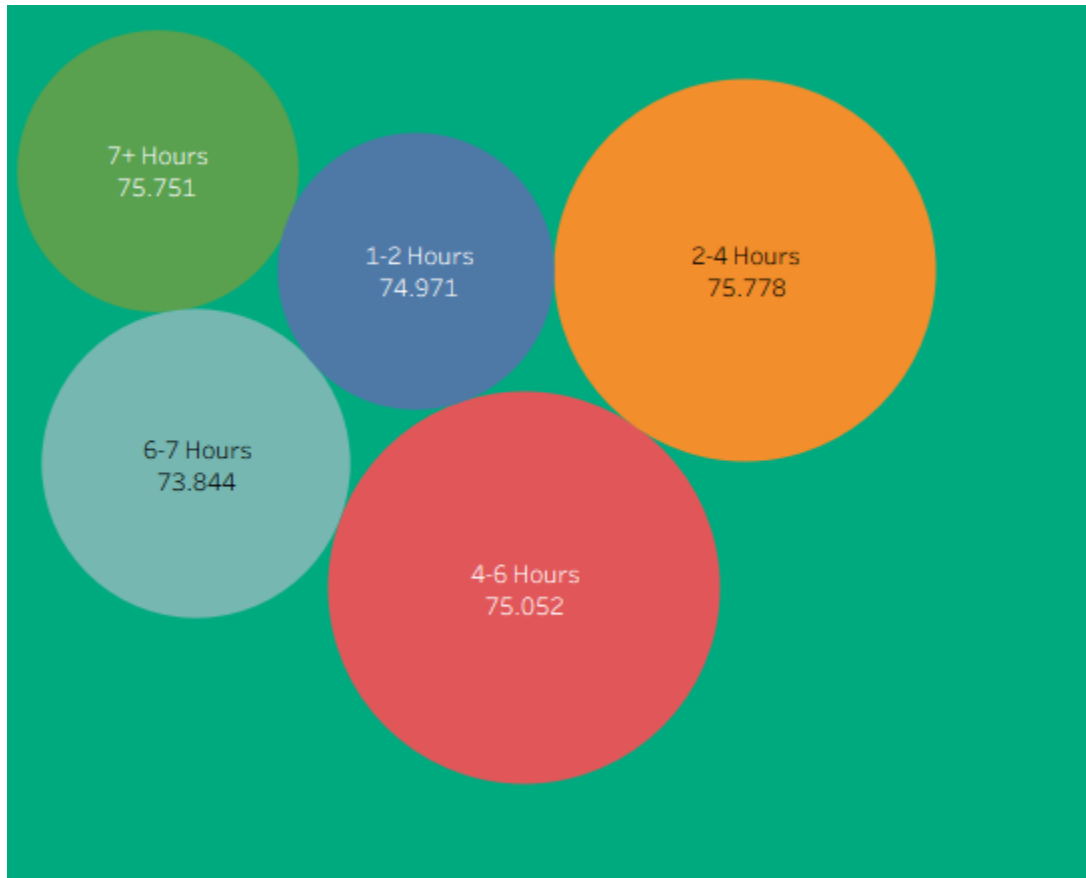
Business Question and Visualization Report

Date	
Team ID	Kalash suhag
Project Name	Analysing mental health in student ecosystem
Maximum Marks	5 Marks

Business Questions and Visualizations

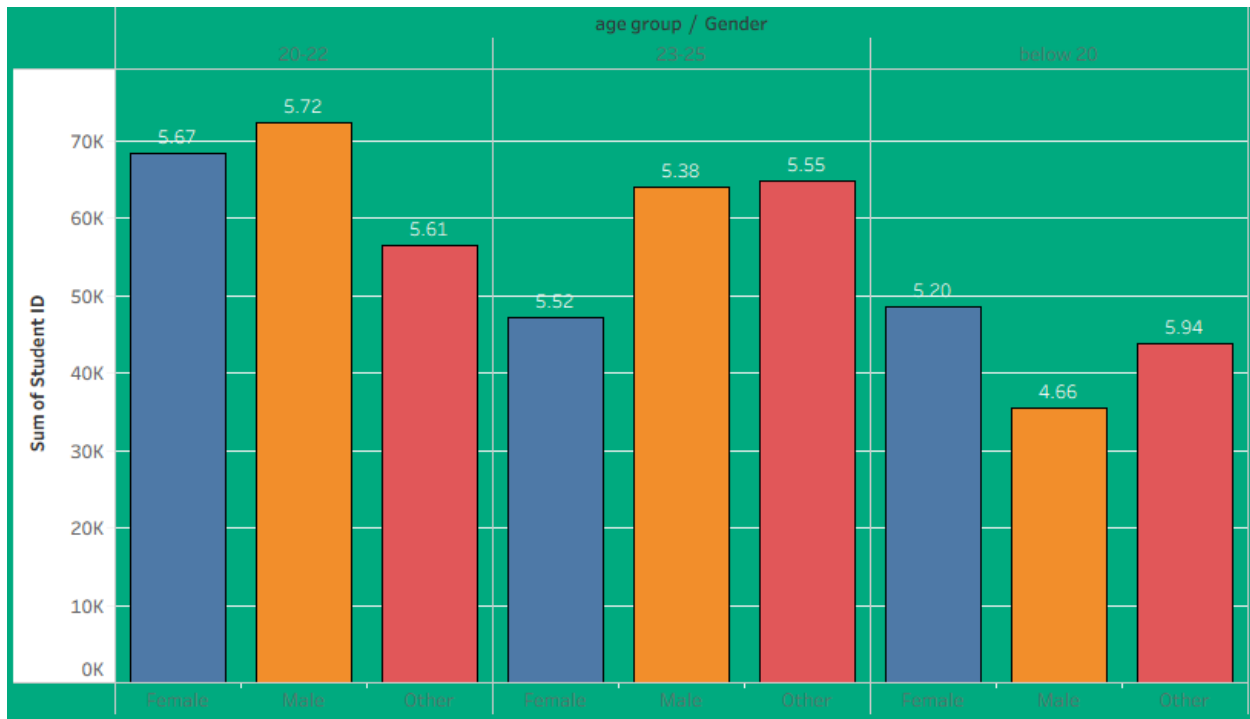
Q1: How do study hours influence students' academic performance?

- **Visualization:** Bubble Chart (Study Hours vs Academic Performance)
- **Insight Provided:** Shows the relationship between the number of hours students study and their academic performance, helping identify whether increased study time is associated with better academic outcomes.



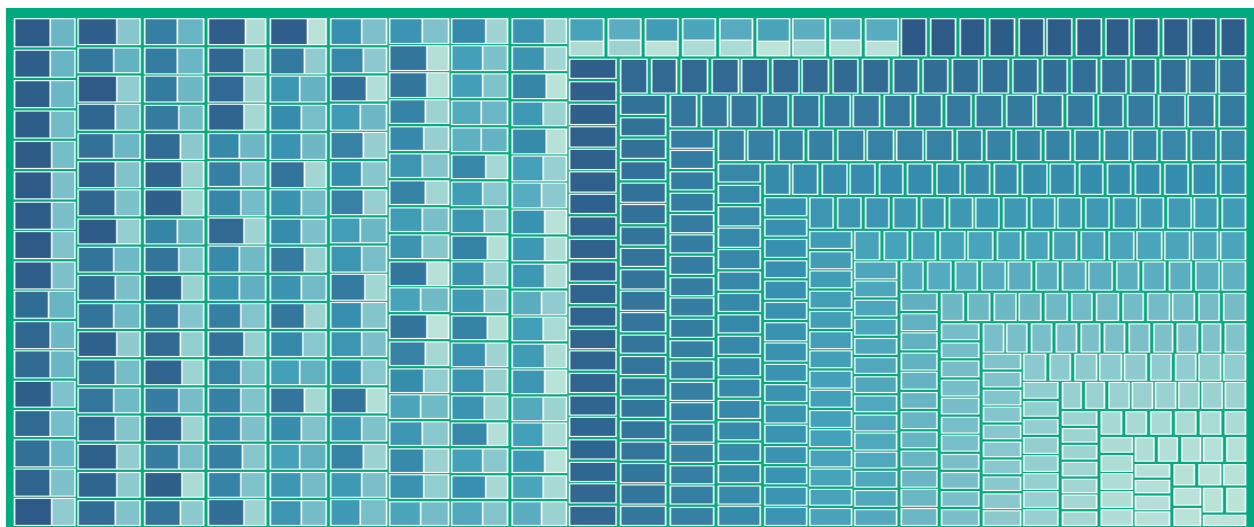
Q2: How does depression level vary across different genders?

- **Visualization:** Bar Chart (Gender vs Depression Level)
- **Insight Provided:** Compares average depression levels among male and female students, enabling counselors to identify gender-based mental health trends.



Q3: What is the relationship between study hours and academic performance across different student groups?

- **Visualization:** Treemap (Study Hours vs Academic Performance)
- **Insight Provided:** Displays how academic performance is distributed across different study-hour categories, helping identify which study patterns are associated with higher performance.



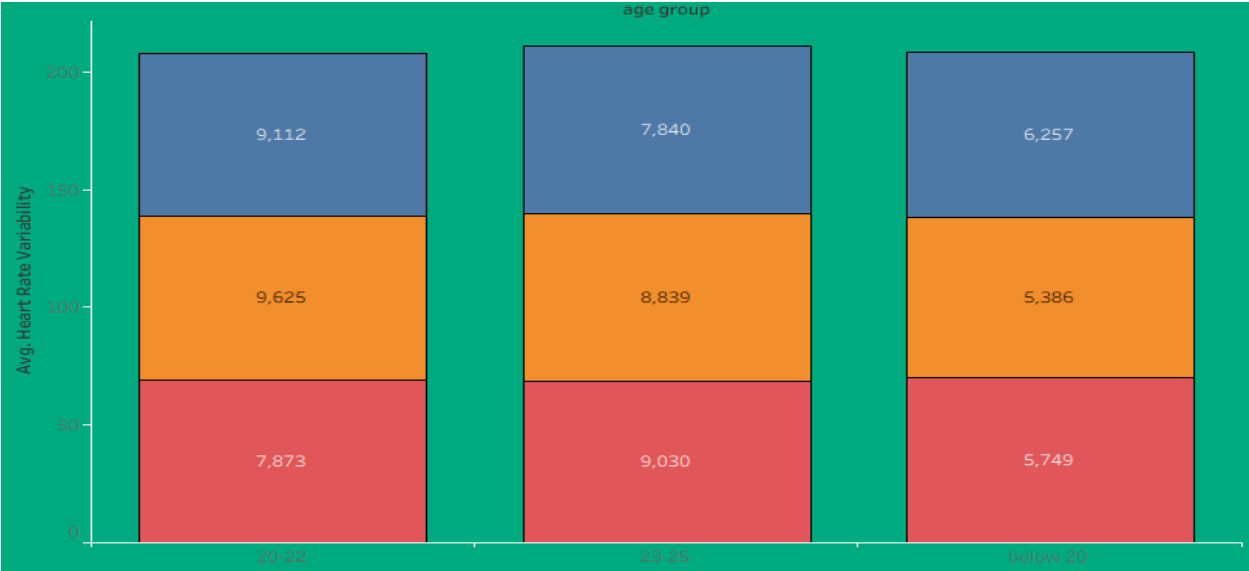
Q4: What is the average stress level among different genders?

- **Visualization:** Lollipop Chart (Average Stress Level by Gender)
- **Insight Provided:** Highlights differences in average stress levels between male and female students, allowing educators to recognize groups experiencing higher stress.



Q5: How does heart rate variability differ by age and gender?

- **Visualization:** Bar Chart (Average Heart Rate Variability by Age and Gender)
- **Insight Provided:** Illustrates variations in heart rate variability across age groups and genders, providing insights into students' physiological well-being.



Q6: What are the key performance indicators (KPIs) related to students' mental health?

Visualization: KPI Cards

Insight Provided:Provides a quick overview of important metrics such as:

- Average Study Hours
- Average Academic Performance
- Average Stress Level
- Average Depression Level
- Total Number of Students

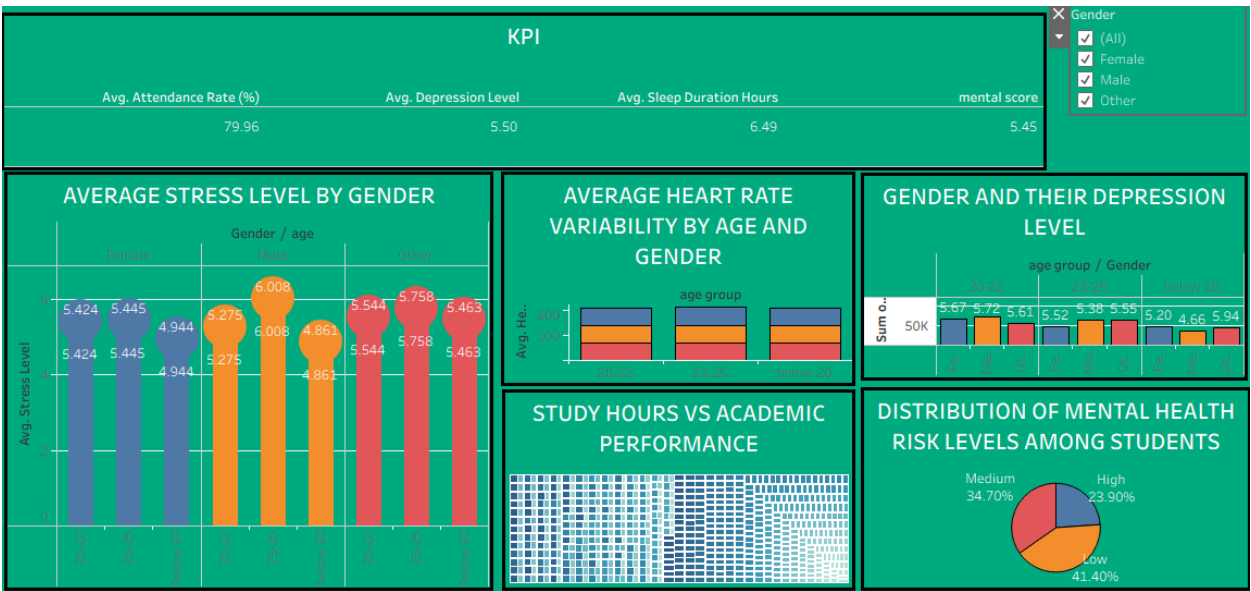
This enables users to assess the overall mental health status of the dataset at a glance.



Q7: How do multiple mental health indicators interact across different student groups?

Visualization: Interactive Dashboard with Filters

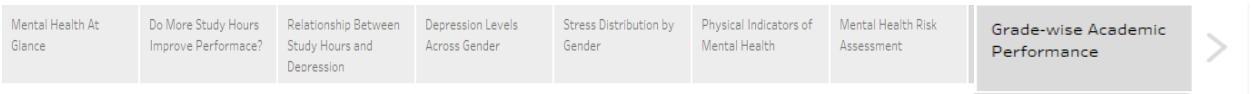
Insight Provided: Allows users to filter data by age, gender, and study hours to explore relationships between stress, depression, academic performance, and physiological measures for different student groups.



Q8: What overall insights can be drawn about students' mental health and academic performance?

Visualization: Tableau Story

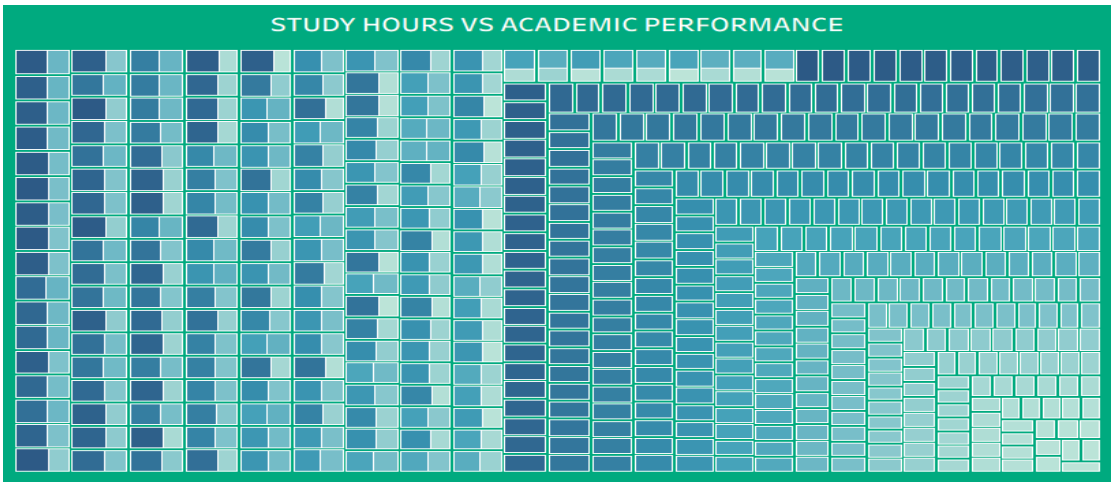
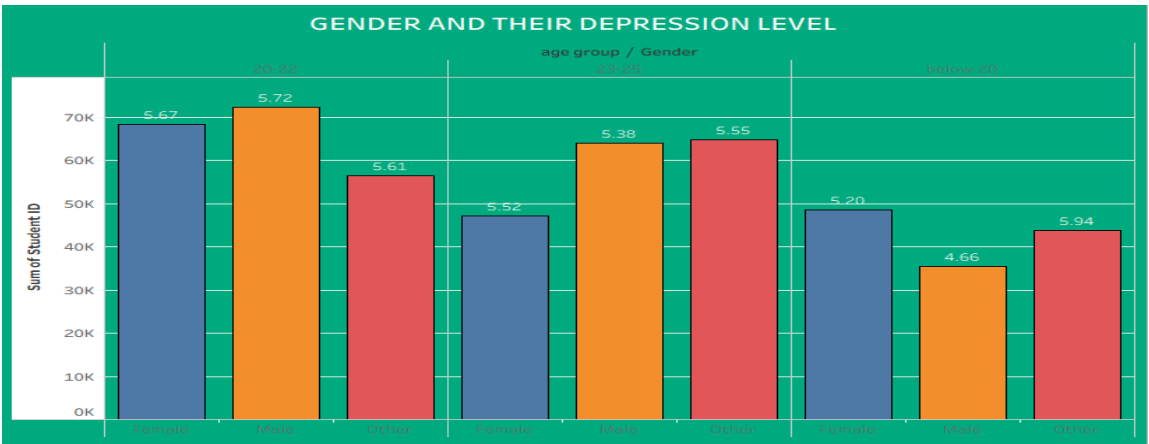
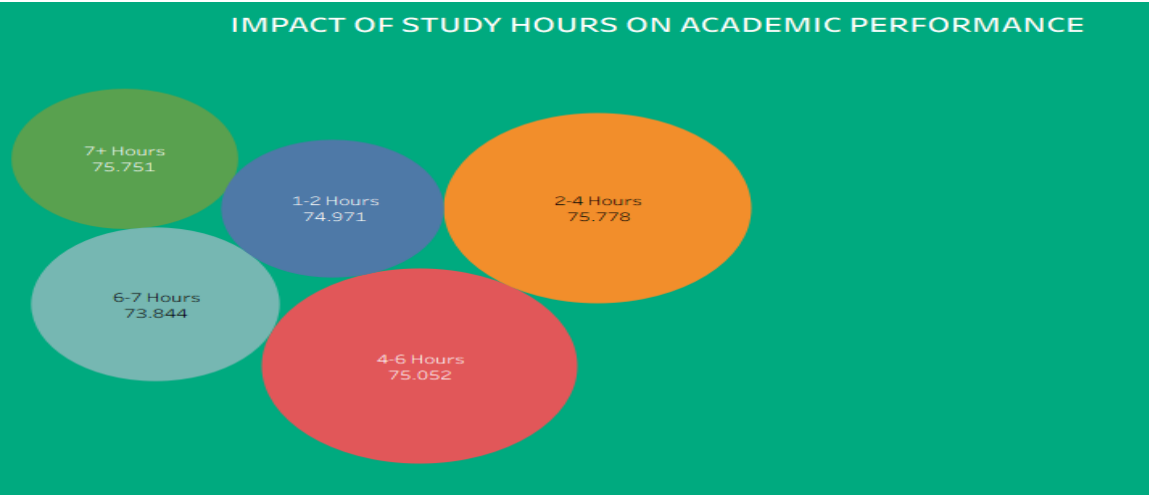
Insight Provided: Combines all dashboard visualizations into a structured narrative, enabling educators and counselors to understand the relationship between study habits, stress, depression, academic performance, and heart rate variability, supporting data-driven decision-making for student well-being.

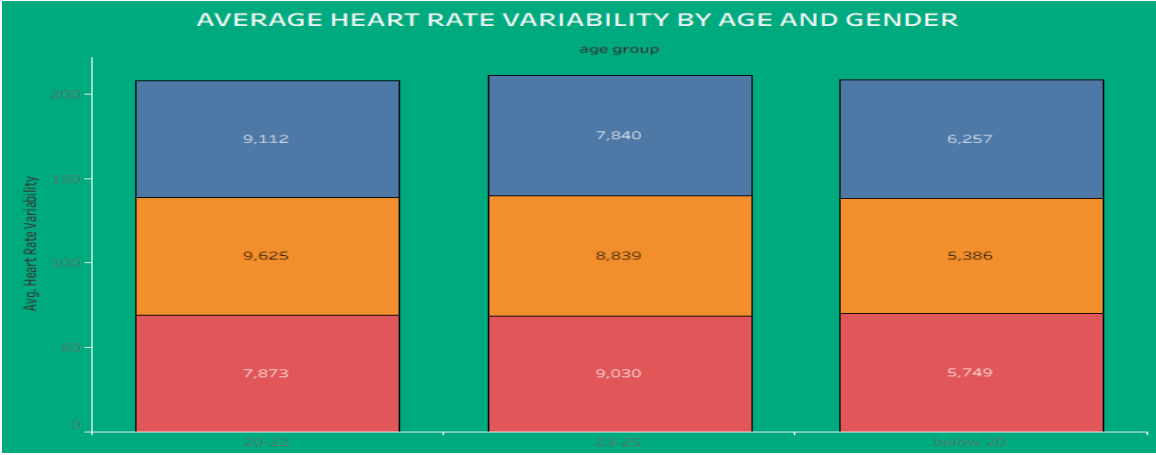
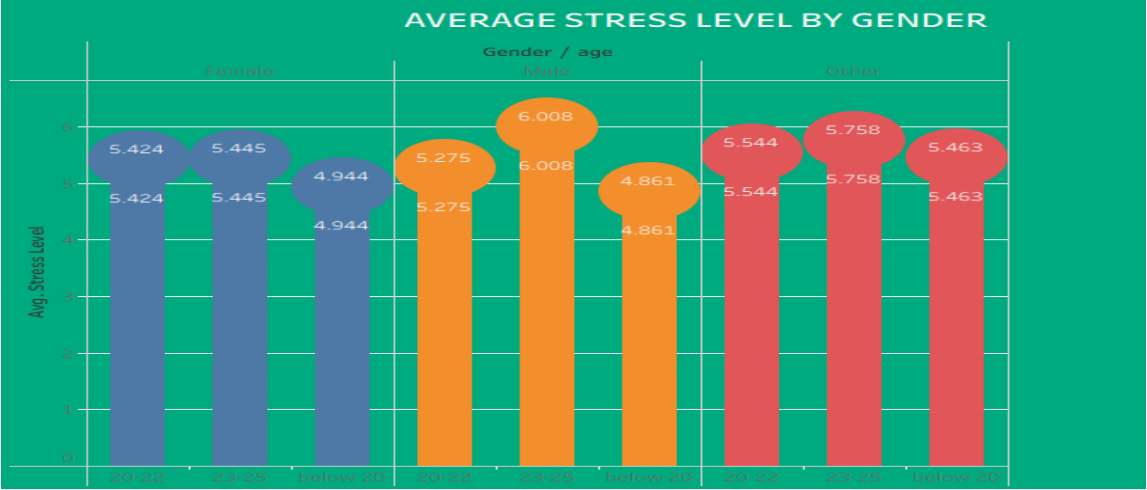


Dashboard Design

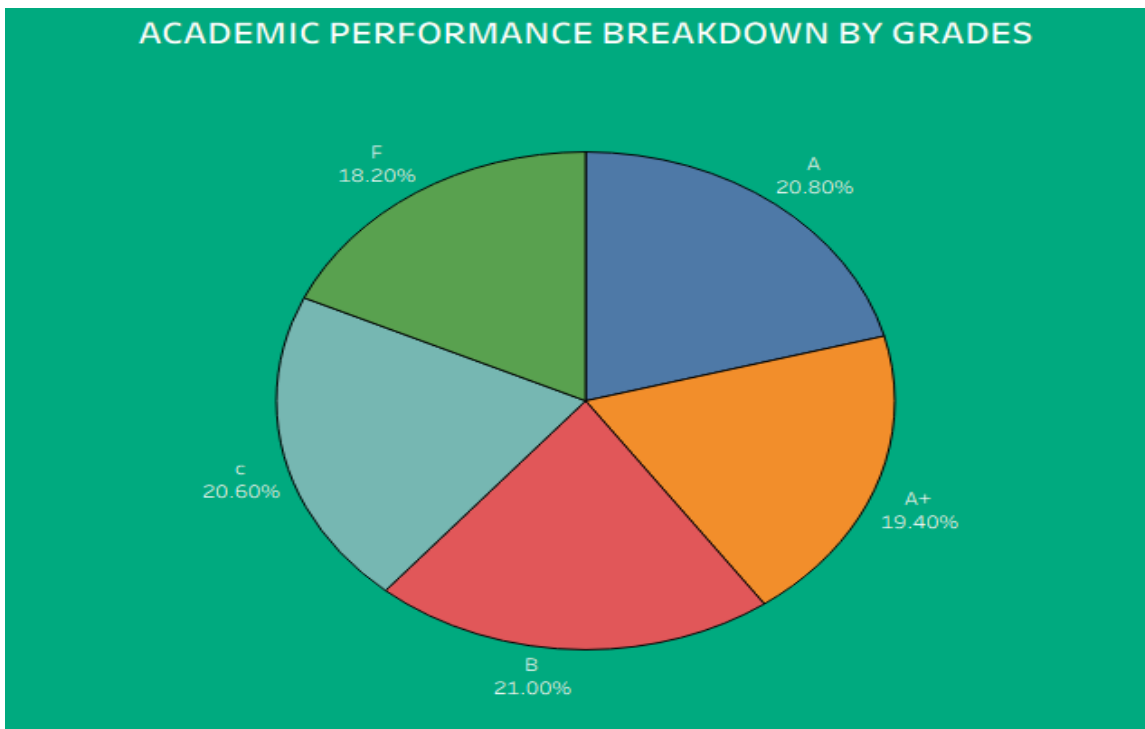
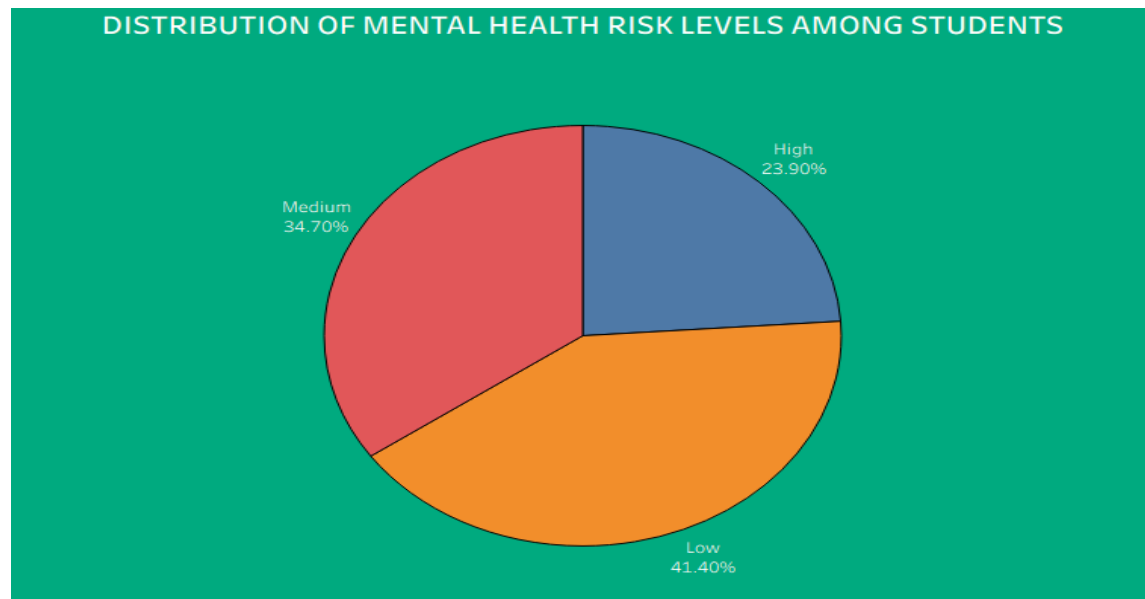
Date	
Team ID	Kalash suhag
Project Name	Analysing mental health in student ecosystem
Maximum Marks	5 Marks

Activity 1: Interactive and Visually Appealing Dashboards





KPI			
Avg. Attendance Rate (%)	Avg. Depression Level	Avg. Sleep Duration Hours	mental score
79.96	5.50	6.49	5.45



The dashboard has been designed by following key visualization principles, including a **clear and intuitive layout, consistent color scheme, interactive filters, responsive dashboard design, and meaningful visual storytelling**. The dashboard enables users to explore relationships between mental health indicators and academic performance efficiently.

The key outcomes and insights presented by the dashboard are:

Study Hours and Academic Performance (Q1)

The **Bubble Chart** demonstrates the relationship between study hours and academic performance. It helps identify how different study durations influence students' academic outcomes and supports better study planning.

Depression Level by Gender (Q2)

The **Bar Chart** compares depression levels across genders, highlighting differences in mental health that can assist educators and counselors in identifying groups requiring additional support.

Study Hours vs Academic Performance (Q3)

The **Treemap** provides a visual comparison of academic performance across different study-hour categories, making it easier to identify study patterns associated with better academic results.

Average Stress Level by Gender (Q4)

The **Lollipop Chart** presents the average stress levels of male and female students, allowing quick comparison of stress distribution between gender groups.

Heart Rate Variability by Age and Gender (Q5)

The visualization illustrates variations in heart rate variability among different age groups and genders, providing insights into students' physiological health alongside mental well-being.

KPI Dashboard

The KPI cards summarize important metrics such as:

- Average Study Hours
- Average Academic Performance
- Average Stress Level
- Average Depression Level
- Total Number of Students

These indicators provide users with a quick overview of the overall dataset.

Interactive Dashboard

Interactive filters for **Age, Gender, and Study Hours** enable users to dynamically explore relationships between academic performance, stress levels, depression, and physiological measures, making the dashboard more user-friendly and insightful.

Activity 2: Live Deployment Link

Below is the verified public access link to the interactive, cloud-hosted visualization story:

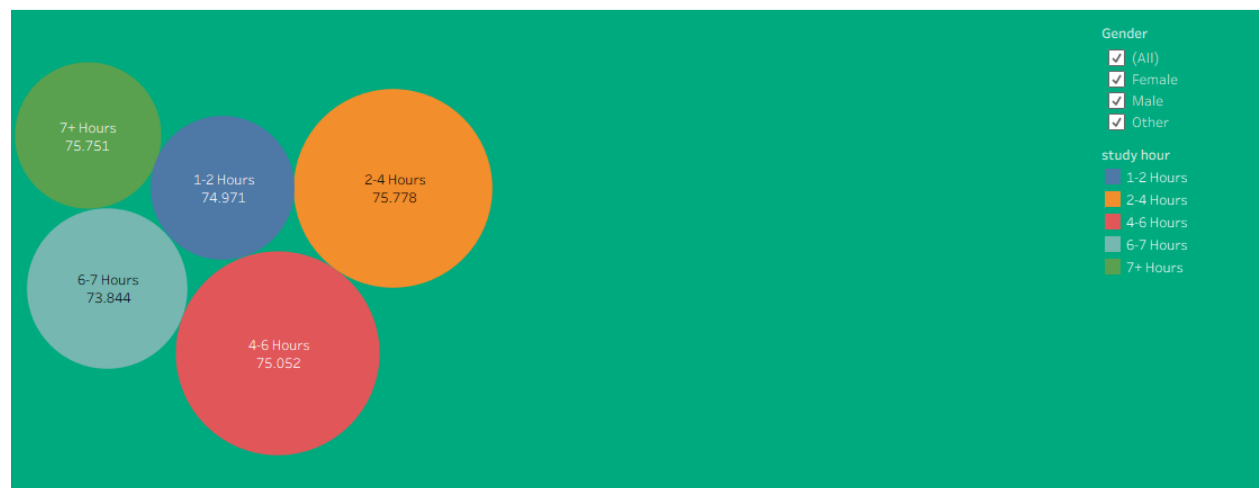
Tableau Public Dashboard Link:

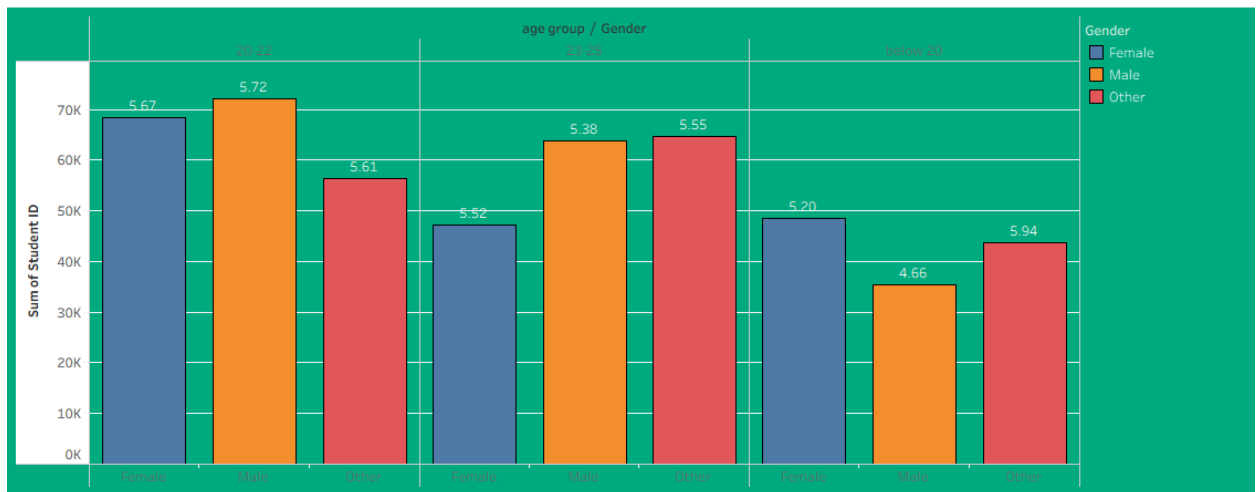
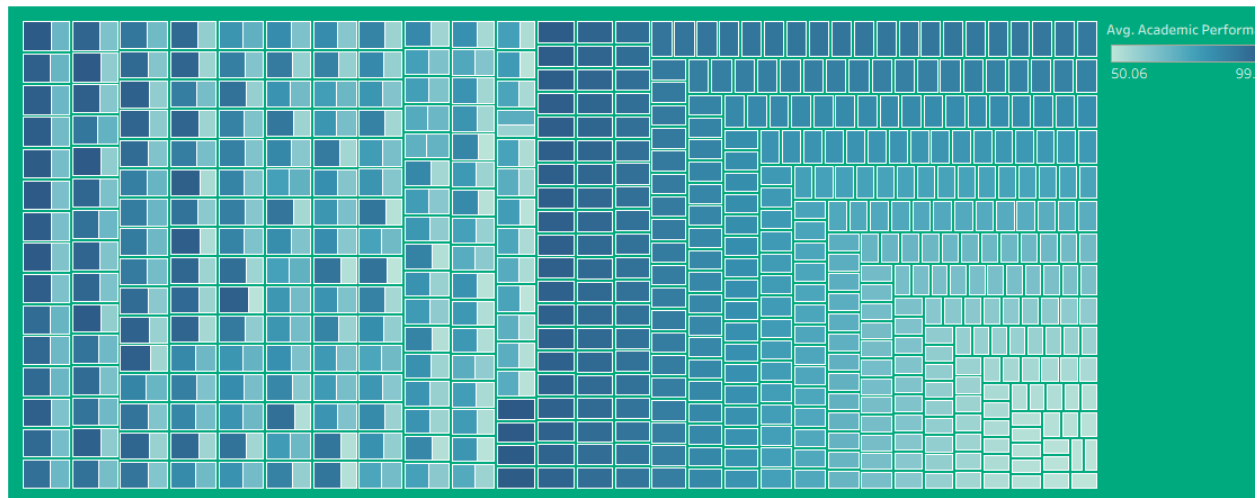
https://public.tableau.com/views/projectworkinternshipdash/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

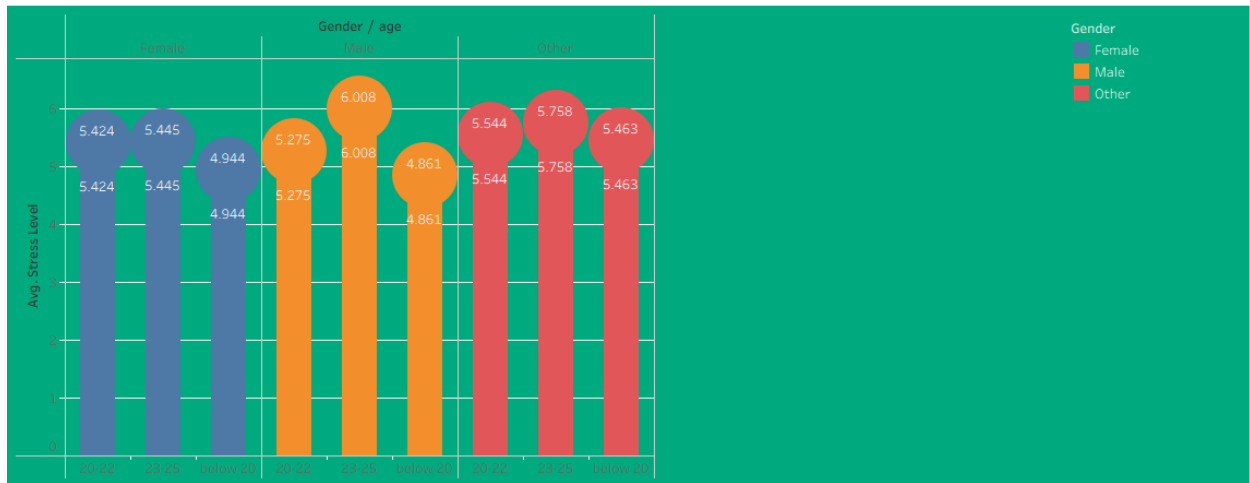
Story

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Activity 1: Story Design:







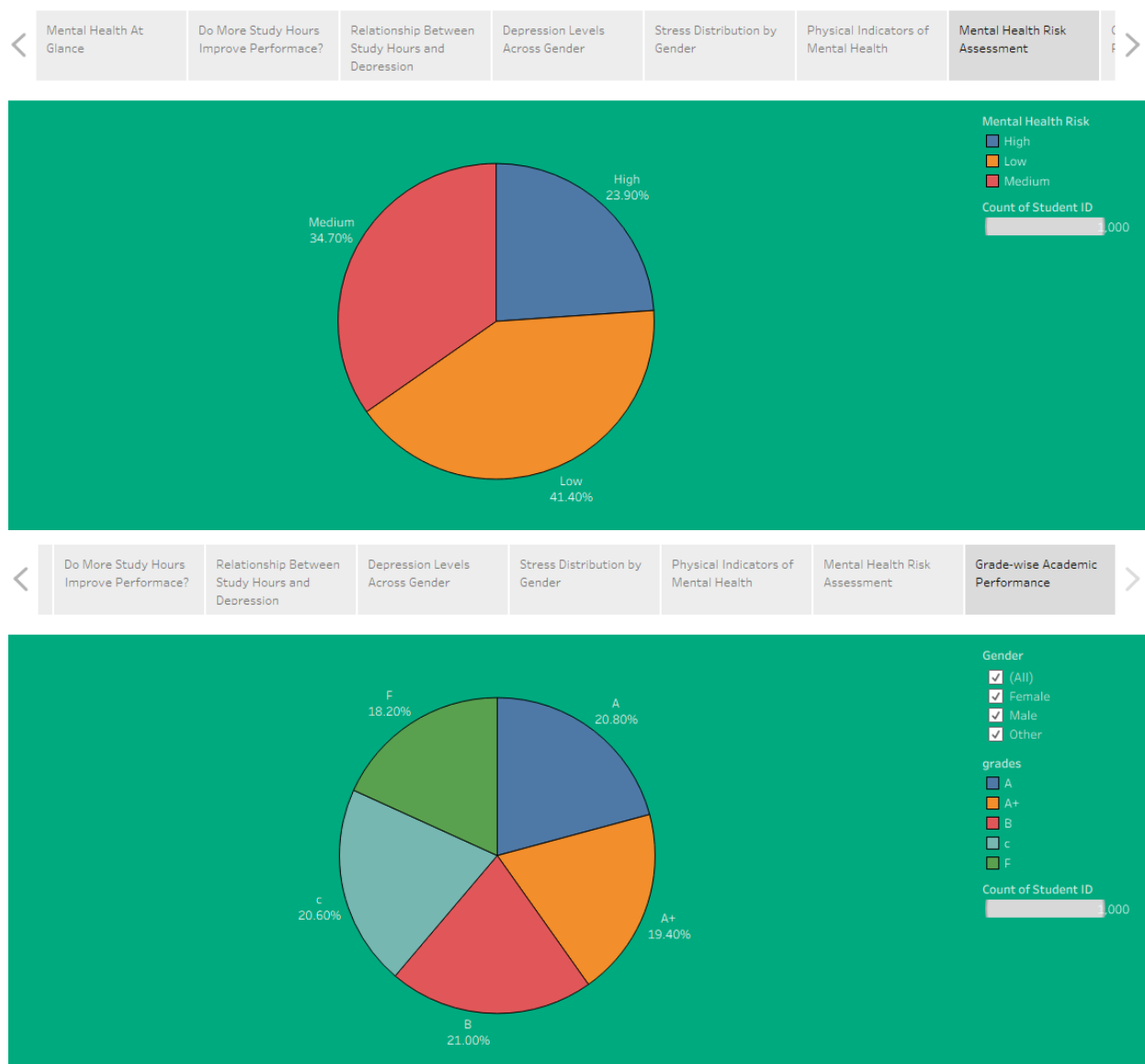


Tableau Story Narrative & Observations

The Tableau Story presents a structured analysis of students' mental health and academic performance by combining multiple visualizations into a logical sequence. The story enables educators, students, and counselors to understand how study habits, stress, depression, and physiological indicators influence academic outcomes.

Based on the interactive story points, the following observations were identified:

1. Study Hours Influence Academic Performance (Q1)

The Bubble Chart shows that students with consistent study hours generally achieve better academic performance. However, excessive or insufficient study hours may not always lead to improved results, indicating the importance of balanced study habits.

2. Depression Levels Differ Across Gender (Q2)

The Bar Chart highlights variations in depression levels between male and female students. This helps identify demographic groups that may require additional mental health support and counseling.

3. Study Habits Affect Academic Outcomes (Q3)

The Treemap demonstrates how academic performance is distributed across different study-hour categories. Students with well-balanced study schedules tend to perform better academically than those with very low study hours.

4. Stress Levels Vary by Gender (Q4)

The Lollipop Chart compares average stress levels between genders, making it easy to identify which group experiences relatively higher stress and may benefit from targeted stress-management initiatives.

5. Physiological Health Supports Mental Health Analysis (Q5)

The Heart Rate Variability visualization illustrates differences across age groups and genders, providing an additional indicator that complements stress and depression analysis.

6. Overall Mental Health Dashboard

The KPI cards and interactive dashboard summarize important metrics such as average study hours, academic performance, stress level, depression level, and total students. Interactive filters allow users to explore relationships among different variables and gain meaningful insights for data-driven decision-making.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/shared/QN3PS4Q8D?:display_count=n&origin=viz_share_link